

SUPERSPACE

or One thousand and one
lessons in supersymmetry

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Contents

Preface

1. Introduction	1
2. A toy superspace	
2.1. Notation and conventions	7
2.2. Supersymmetry and superfields	9
2.3. Scalar multiplet	15
2.4. Vector multiplet	18
2.5. Other global gauge multiplets	28
2.6. Supergravity	34
2.7. Quantum superspace	46
3. Representations of supersymmetry	
3.1. Notation	54
3.2. The supersymmetry groups	62
3.3. Representations of supersymmetry	69
3.4. Covariant derivatives	83
3.5. Constrained superfields	89
3.6. Component expansions	92
3.7. Superintegration	97
3.8. Superfunctional differentiation and integration	101
3.9. Physical, auxiliary, and gauge components	108
3.10. Compensators	112
3.11. Projection operators	120
3.12. On-shell representations and superfields	138
3.13. Off-shell field strengths and prepotentials	147
4. Classical, global, simple ($N = 1$) superfields	
4.1. The scalar multiplet	149
4.2. Yang-Mills gauge theories	159
4.3. Gauge-invariant models	178
4.4. Superforms	181
4.5. Other gauge multiplets	198
4.6. N -extended multiplets	216
5. Classical $N = 1$ supergravity	
5.1. Review of gravity	232
5.2. Prepotentials	244

$$= iL_V [\cosh(\frac{1}{2} L_V)(\bar{\Lambda} - \Lambda) - \sinh(\frac{1}{2} L_V)(\bar{\Lambda} + \Lambda)] , \quad (4.2.27)$$

from which it follows

$$\begin{aligned} \delta V &= -\frac{1}{2} iL_V [\bar{\Lambda} + \Lambda + \coth(\frac{1}{2} L_V)(\Lambda - \bar{\Lambda})] \\ &= i(\bar{\Lambda} - \Lambda) - \frac{1}{2} i[V, \bar{\Lambda} + \Lambda] + O(V^2) . \end{aligned} \quad (4.2.28)$$

From the transformations (4.2.28) and the parameter (4.2.11-12) we find the non-abelian WZ gauge-preserving “supersymmetry” transformations:

$$\begin{aligned} \delta A_{\underline{a}} &= -i(\bar{\epsilon}_{\dot{\alpha}} \lambda_{\alpha} + \epsilon_{\alpha} \bar{\lambda}_{\dot{\alpha}}) , \\ \delta \lambda_{\alpha} &= -\epsilon^{\beta} f_{\beta\alpha} + i\epsilon_{\alpha} D' , \\ \delta D' &= \frac{1}{2} \nabla_{\alpha\dot{\alpha}} (\bar{\epsilon}^{\dot{\alpha}} \lambda^{\alpha} - \epsilon^{\alpha} \bar{\lambda}_{\dot{\alpha}}) , \end{aligned} \quad (4.2.29)$$

where now $f_{\alpha\beta}$ is the self-dual part of the *nonabelian* field strength and $\nabla_{\alpha\dot{\alpha}} = \partial_{\alpha\dot{\alpha}} - iA_{\alpha\dot{\alpha}}$. The nonlinearity comes from the gauge-covariantization of the linear transformations (4.2.13). The components of the nonabelian vector multiplet are covariant generalizations of the abelian components; in the WZ gauge, they are the same as (4.2.4a) (see also (4.3.5)).

a.3. Covariant derivatives

The gauge field V can be used to construct derivatives, gauge covariant *with respect to Λ transformations*

$$\nabla_A = D_A - i\Gamma_A = (\nabla_{\alpha}, \nabla_{\dot{\alpha}}, \nabla_{\alpha\dot{\alpha}}) , \quad (4.2.30)$$

defined by the requirement

$$(\nabla_A \Phi)' = e^{i\Lambda} (\nabla_A \Phi) , \quad (4.2.31)$$

i.e.,

$$\nabla'_A = e^{i\Lambda} \nabla_A e^{-i\Lambda} , \quad (4.2.32a)$$

or

$$\delta \nabla_A = i[\Lambda, \nabla_A] \quad . \quad (4.2.32b)$$

Since Λ is chiral, $\nabla_{\dot{\alpha}} \equiv \bar{D}_{\dot{\alpha}}$ is covariant without further modification:

$$\nabla'_{\dot{\alpha}} = e^{i\Lambda} \bar{D}_{\dot{\alpha}} e^{-i\Lambda} = \nabla_{\dot{\alpha}} \quad . \quad (4.2.33)$$

The undotted spinor derivative D_{α} is covariant with respect to $\bar{\Lambda}$ transformations. We can use e^V to convert it into a derivative covariant with respect to Λ (see (4.2.21)); $\nabla_{\alpha} \equiv e^{-V} D_{\alpha} e^V$ transforms correctly:

$$\begin{aligned} \nabla'_{\alpha} &= (e^{i\Lambda} e^{-V} e^{-i\bar{\Lambda}}) D_{\alpha} (e^{i\bar{\Lambda}} e^V e^{-i\Lambda}) \\ &= e^{i\Lambda} e^{-V} D_{\alpha} e^V e^{-i\Lambda} \\ &= e^{i\Lambda} \nabla_{\alpha} e^{-i\Lambda} \quad . \end{aligned} \quad (4.2.34)$$

Finally, we construct $\nabla_{\underline{a}}$ by analogy with (3.4.9): $\nabla_{\underline{a}} = \nabla_{\alpha\dot{\alpha}} \equiv -i\{\nabla_{\alpha}, \nabla_{\dot{\alpha}}\}$. Its covariance follows from that of ∇_{α} and $\nabla_{\alpha\dot{\alpha}}$.

We summarize:

$$\nabla_A = (e^{-V} D_{\alpha} e^V, \bar{D}_{\dot{\alpha}}, -i\{\nabla_{\alpha}, \nabla_{\dot{\alpha}}\}) \quad . \quad (4.2.35)$$

These derivatives are not hermitian. Their conjugates $\bar{\nabla}_A$ are covariant with respect to $\bar{\Lambda}$ transformations:

$$\begin{aligned} \bar{\nabla}_A &= (D_{\alpha}, e^V \bar{D}_{\dot{\alpha}} e^{-V}, -i\{\bar{\nabla}_{\alpha}, \bar{\nabla}_{\dot{\alpha}}\}) \quad , \\ \bar{\nabla}'_A &= e^{i\bar{\Lambda}} \bar{\nabla}_A e^{-i\bar{\Lambda}} \quad . \end{aligned} \quad (4.2.36)$$

The derivatives ∇_A ($\bar{\nabla}_A$) are called *gauge chiral* (*antichiral*) *representation* covariant derivatives. They are related by a nonunitary similarity transformation

$$\bar{\nabla}_A = e^V \nabla_A e^{-V} \quad . \quad (4.2.37)$$

This is analogous to the relation between global supersymmetry chiral and antichiral representations

$$D_A^{(-)} = e^U D_A^{(+)} e^{-U} \quad (4.2.38)$$

of (3.4.8).

The gauge covariant derivatives are usually defined in terms of vector representation D_A 's; if we express these in terms of ordinary derivatives, (4.2.35) becomes

$$\nabla_A = (e^{-V} e^{-\frac{1}{2}U} \partial_\alpha e^{\frac{1}{2}U} e^V, e^{\frac{1}{2}U} \bar{\partial}_{\dot{\alpha}} e^{-\frac{1}{2}U}, -i\{\nabla_\alpha, \nabla_{\dot{\alpha}}\}) \quad . \quad (4.2.39)$$

By a further similarity transformation $\nabla_A \rightarrow e^{-\frac{1}{2}U} \nabla_A e^{\frac{1}{2}U}$, we go to a new representation that is chiral with respect to both global supersymmetry and gauge transformations:

$$\nabla_A = (e^{-\frac{1}{2}U} e^{-V} e^{-\frac{1}{2}U} \partial_\alpha e^{\frac{1}{2}U} e^V e^{\frac{1}{2}U}, \bar{\partial}_{\dot{\alpha}}, -i\{\nabla_\alpha, \nabla_{\dot{\alpha}}\}) \quad . \quad (4.2.40)$$

We define $\tilde{V}$ by

$$e^{\frac{1}{2}U} e^V e^{\frac{1}{2}U} = e^{U+\tilde{V}} \quad . \quad (4.2.41)$$

In this form, it is clear that $\tilde{V}$ gauge covariantizes U : $i\theta^\alpha \bar{\theta}^{\dot{\alpha}} \partial_{\alpha\dot{\alpha}} \rightarrow \dots + i\theta^\alpha \bar{\theta}^{\dot{\alpha}} (\partial_{\alpha\dot{\alpha}} - iA_{\alpha\dot{\alpha}}) + \dots$. This combination transforms as

$$(e^{U+\tilde{V}})' = e^{i\bar{\Lambda}} (e^{U+\tilde{V}}) e^{-i\Lambda} \quad , \quad \partial_\alpha \bar{\Lambda} = \bar{\partial}_{\dot{\alpha}} \Lambda = 0 \quad . \quad (4.2.42)$$

There also exists a symmetric *gauge vector representation* that treats chiral and antichiral fields on the same footing. Such a representation uses a complex scalar gauge field Ω , and requires a larger gauge group. We discuss the vector representation in subsec. 4.2.b, where the covariant derivatives are defined abstractly, and where it enters naturally.

a.4. Field strengths

The covariant derivatives define field strengths by commutation:

$$[\nabla_A, \nabla_B] = T_{AB}{}^C \nabla_C - iF_{AB} \quad , \quad (4.2.43)$$

with $V = V^A T_A$, and T_A in the adjoint representation. From the explicit form of the covariant derivatives (4.2.35) we find that the torsion $T_{AB}{}^C$ is the same one as in flat global superspace (3.4.19), and some field strengths vanish:

$$F_{\alpha\beta} = F_{\dot{\alpha}\dot{\beta}} = F_{\alpha\dot{\beta}} = 0 \quad . \quad (4.2.44)$$

The remaining field strengths are

$$F_{\dot{\alpha},\beta\dot{\beta}} = C_{\dot{\alpha}\beta} \bar{D}^2 (e^{-V} D_\beta e^V) = iC_{\dot{\alpha}\beta} W_\beta \quad ,$$

$$F_{\alpha\dot{\alpha},\beta\dot{\beta}} = \frac{1}{2} (C_{\dot{\alpha}\dot{\beta}} \nabla_{(\alpha} W_{\beta)} + C_{\alpha\beta} \nabla_{(\dot{\alpha}} W_{\dot{\beta}})) ,$$

$$W_{\alpha} \equiv i \bar{D}^2 (e^{-V} D_{\alpha} e^V) ,$$

$$W_{\dot{\alpha}} \equiv e^{-V} \bar{W}_{\dot{\alpha}} e^V \equiv e^{-V} (-W_{\alpha})^{\dagger} e^V . \quad (4.2.45)$$

(Recall that $\bar{W}^{\dot{\alpha}} \equiv (W^{\alpha})^{\dagger}$ implies $\bar{W}_{\dot{\alpha}} = (-W_{\alpha})^{\dagger}$ (3.1.20).) Thus all the field strengths of the theory are expressed in terms of a single spinor W_{α} that is the nonlinear version of (4.2.2). It satisfies Bianchi identities analogous to (4.2.1):

$$\nabla^{\alpha} W_{\alpha} = - \nabla^{\dot{\alpha}} W_{\dot{\alpha}} . \quad (4.2.46)$$

It is chiral, has dimension $\frac{3}{2}$, and can be used to construct a gauge invariant action

$$S = \frac{1}{g^2} \text{tr} \int d^4x d^2\theta W^2 = - \frac{1}{2g^2} \text{tr} \int d^4x d^4\theta (e^{-V} D^{\alpha} e^V) \bar{D}^2 (e^{-V} D_{\alpha} e^V) ,$$

$$V = V^A T_A , \quad \text{tr} T_A T_B = \delta_{AB} . \quad (4.2.47)$$

As in the abelian case, this action is hermitian up to a surface term (see discussion following (4.2.15)).

a.5. Covariant variations

To derive the field equations from the action (4.2.47), we need to vary the action with respect to V . However, since V is not a covariant object, this results in noncovariant field equations (although multiplication by a suitable (but complicated) invertible operator covariantizes them). In addition, variation with respect to V is complicated because V appears in e^V factors. We therefore define a covariant variation of V by

$$\Delta V \equiv e^{-V} \delta e^V = \frac{1 - e^{-L_V}}{L_V} \delta V = \delta V + \dots . \quad (4.2.48)$$

ΔV satisfies the chiral representation hermiticity condition as in (4.2.37). In practice, we always vary an action with respect to V by expressing its variation in terms of δe^V , and then rewriting that in terms of ΔV . We thus define a covariant functional derivative $\frac{\Delta F[V]}{\Delta V}$ by (cf. (3.8.3))

b. Covariant approach

In this subsection we discuss another approach to supersymmetric Yang-Mills theory that reverses the direction of the previous section. We postulate derivatives transforming covariantly under a gauge group, impose constraints on them, and discover that they can be expressed in terms of prepotentials. This procedure will prove especially useful in studying supergravity and extended super-Yang-Mills, so we give a detailed analysis for the simpler case of $N = 1$ super-Yang-Mills.

We start with the ordinary superspace derivatives D_A satisfying $[D_A, D_B] = T_{AB}{}^C D_C$, where $T_{AB}{}^C$ is the torsion and has only one nonzero component $T_{\alpha\dot{\beta}}{}^{\dot{c}}$ (see(3.4.19)). For a Lie algebra with generators T_A we covariantize the derivatives by introducing connection fields

$$\nabla_A = D_A - i\Gamma_A \quad , \quad (4.2.53)$$

where $\Gamma_A = \Gamma_A{}^B T_B$ is hermitian and $\bar{\nabla}_A = -(-)^A \nabla_A$. At the component level we have

$$\Gamma_\alpha = v_\alpha + \frac{i}{2} \bar{\theta}^{\dot{\alpha}} v_{\alpha\dot{\alpha}} + \cdots \quad , \quad \Gamma_{\underline{a}} = w_{\underline{a}} + \cdots \quad , \quad (4.2.54a)$$

and hence

$$\nabla_\alpha = \partial_\alpha - iv_\alpha + \frac{i}{2} \bar{\theta}^{\dot{\alpha}} (\partial_{\alpha\dot{\alpha}} - iv_{\alpha\dot{\alpha}}) + \cdots \quad ,$$

$$\nabla_{\dot{\alpha}} = \bar{\partial}_{\dot{\alpha}} - i\bar{v}_{\dot{\alpha}} + \frac{i}{2} \theta^\alpha (\partial_{\alpha\dot{\alpha}} - i\bar{v}_{\alpha\dot{\alpha}}) + \cdots \quad ,$$

$$\nabla_a = \partial_a - iw_a + \cdots \quad , \quad (4.2.54b)$$

so that the component derivatives are covariantized.

Under gauge transformations the covariant derivatives are postulated to transform as

$$\nabla'_A = e^{iK} \nabla_A e^{-iK} \quad , \quad (4.2.55)$$

where the parameter $K = K^A T_A$ is a real superfield.

$$K = \omega(x) + \theta^\alpha K^{(1)}_\alpha(x) + \bar{\theta}^{\dot{\alpha}} \bar{K}^{(1)}_{\dot{\alpha}}(x) + \cdots \quad . \quad (4.2.56)$$

This is very different from what emerged in the previous section: Instead of chiral

representation derivatives transforming with the chiral parameter Λ , we have *vector representation* hermitian derivatives, transforming with the hermitian parameter K . The asymmetric form of the previous section will emerge when we make a similarity transformation to go to the chiral representation.

For infinitesimal K , we find the component transformations:

$$\begin{aligned}\delta v_{\alpha\dot{\alpha}} &= [\partial_{\alpha\dot{\alpha}} - i v_{\alpha\dot{\alpha}}, \omega] - i \omega_{\alpha\dot{\alpha}} \quad , \\ \delta w_{\alpha\dot{\alpha}} &= [\partial_{\alpha\dot{\alpha}} - i w_{\alpha\dot{\alpha}}, \omega] \quad ,\end{aligned}\tag{4.2.57}$$

where $\omega \equiv K|$, $\omega_{\alpha\dot{\alpha}} \equiv [\bar{D}_{\dot{\alpha}}, D_{\alpha}]K| = \bar{\omega}_{\alpha\dot{\alpha}}$. The component gauge parameter $\omega_{\alpha\dot{\alpha}}$ can be used to gauge away $Im v_{\alpha\dot{\alpha}}$ algebraically; however, the component fields $Re v_{\alpha\dot{\alpha}}$ and $w_{\alpha\dot{\alpha}}$ both remain as two a priori independent gauge fields for the *same* component gauge transformation. To avoid this we impose *constraints* on the covariant derivatives.

b.1. Conventional constraints

Field strengths F_{AB} are defined by (4.2.43). Substituting (4.2.53) we find

$$F_{AB} = D_{[A}\Gamma_{B]} - i[\Gamma_A, \Gamma_B] - T_{AB}{}^C \Gamma_C \quad .\tag{4.2.58}$$

In particular,

$$F_{\alpha\dot{\alpha}} = D_{\alpha}\bar{\Gamma}_{\dot{\alpha}} + \bar{D}_{\dot{\alpha}}\Gamma_{\alpha} - i\{\Gamma_{\alpha}, \bar{\Gamma}_{\dot{\alpha}}\} - i\Gamma_{\alpha\dot{\alpha}} \quad .\tag{4.2.59}$$

If we impose the constraint

$$F_{\alpha\dot{\alpha}} = 0 \quad ,\tag{4.2.60}$$

(4.2.59) *defines* the vector connection $\Gamma_{\alpha\dot{\alpha}}$ in terms of the spinor connections. (In components, this expresses $w_{\alpha\dot{\alpha}}$ in terms of $v_{\alpha\dot{\alpha}}$ and $v_{\alpha\dot{\alpha}}$.)

In any theory one can add covariant terms to the connections (e.g., (3.10.22)) without changing the transformation of the covariant derivatives. If we did not impose the constraint (4.2.60) on the connections Γ_A , we could define equally satisfactory new connections $\Gamma'_A = (\Gamma_{\alpha}, \bar{\Gamma}_{\dot{\alpha}}, \Gamma_{\alpha\dot{\alpha}} - iF_{\alpha\dot{\alpha}})$ that identically satisfy the constraints. For this reason (4.2.60) is called a *conventional* constraint. It implies

$$\nabla_A = (\nabla_{\alpha}, \bar{\nabla}_{\dot{\alpha}}, -i\{\nabla_{\alpha}, \bar{\nabla}_{\dot{\alpha}}\}) \quad .\tag{4.2.61}$$

The theory now is expressed entirely in terms of the connection Γ_α . However, it contains spin $s > 1$ gauge covariant component fields, for example

$$\psi_{(\alpha\beta)\dot{\beta}} \equiv F_{(\alpha,\beta)\dot{\beta}}| = i[\bar{D}_{\dot{\beta}}, D_{(\alpha}\Gamma_{\beta)}]| + \cdots \quad (4.2.62)$$

It also contains a superfield strength $F_{\alpha\beta}$ whose θ -independent component

$$f_{\alpha\beta} = F_{\alpha\beta}| = D_{(\alpha}\Gamma_{\beta)}| + \cdots \quad (4.2.63)$$

is a dimension one symmetric spinor (equivalent to an antisymmetric second rank tensor). Because of its dimension, it cannot be the Yang-Mills field strength. Although in principle the theory might contain such fields (as auxiliary, not physical, components), in the covariant approach there are generally further types of constraints that eliminate (many) such components.

b.2. Representation-preserving constraints

To couple scalar multiplets described by chiral scalar superfields to super-Yang-Mills theory, we must define *covariantly chiral* superfields Φ : The covariant derivatives transform with the hermitian parameter K , and all fields must either be neutral or transform with the same parameter. However, K is not chiral, and gauge transformations will not preserve chirality defined with $\bar{D}_{\dot{\alpha}}$. Instead we define a covariantly chiral superfield by

$$\begin{aligned} \bar{\nabla}_{\dot{\alpha}}\Phi &= 0 \quad , \quad \Phi' = e^{iK}\Phi \quad , \\ \nabla_{\alpha}\bar{\Phi} &= 0 \quad , \quad \bar{\Phi}' = \bar{\Phi}e^{-iK} \quad . \end{aligned} \quad (4.2.64)$$

This implies

$$0 = \{\bar{\nabla}_{\dot{\alpha}}, \bar{\nabla}_{\dot{\beta}}\}\Phi = -iF_{\dot{\alpha}\dot{\beta}}\Phi \quad (4.2.65)$$

Consistency requires that we impose the *representation-preserving* constraint

$$F_{\alpha\beta} = \bar{F}_{\dot{\alpha}\dot{\beta}} = 0 \quad (4.2.66)$$

This can be written as

$$\{\nabla_{\alpha}, \nabla_{\beta}\} = 0 \quad (4.2.67)$$

The most general solution is

$$\nabla_\alpha = e^{-\Omega} D_\alpha e^\Omega \quad , \quad \Omega = \Omega^A T_A \quad , \quad (4.2.68)$$

where Ω^A is an arbitrary *complex* superfield. Eq. (4.2.67) states that ∇_α satisfies the same algebra as D_α , and the solution expresses the fact that they are equivalent up to a *complex* gauge transformation. Hermitian conjugation yields

$$\bar{\nabla}_{\dot{\alpha}} = e^{\bar{\Omega}} \bar{D}_{\dot{\alpha}} e^{-\bar{\Omega}} \quad . \quad (4.2.69)$$

Thus ∇_A is completely expressed in terms of the unconstrained *prepotential* Ω by the solutions (4.2.61,68,69) to the constraints (4.2.60,66).

The K gauge transformations are realized by

$$(e^\Omega)' = e^\Omega e^{-iK} \quad . \quad (4.2.70)$$

However, the solution to the constraint (4.2.67) has introduced an additional gauge invariance: The covariant derivatives (4.2.68) are *invariant* under the transformation

$$(e^\Omega)' = e^{i\bar{\Lambda}} e^\Omega \quad , \quad \bar{D}_{\dot{\alpha}} \Lambda = 0 \quad . \quad (4.2.71)$$

Therefore, the gauge group of Ω is larger than that of Γ_A .

We define the K -invariant hermitian part of Ω by

$$e^V = e^\Omega e^{\bar{\Omega}} \quad . \quad (4.2.72)$$

The K gauge transformations can be used to gauge away the antihermitian part of Ω . In this gauge, $\Omega = \bar{\Omega} = \frac{1}{2} V$, and Λ transformations must be accompanied by gauge-restoring K transformations:

$$(e^\Omega)' = e^{i\bar{\Lambda}} e^\Omega e^{-iK(\Lambda)} \quad ,$$

$$e^{-iK(\Lambda)} = e^{-\Omega} e^{-i\bar{\Lambda}} (e^{i\bar{\Lambda}} e^{2\Omega} e^{-i\Lambda})^{\frac{1}{2}} \quad . \quad (4.2.73)$$

In *any* gauge, the transformation of V is

$$(e^V)' = e^{i\bar{\Lambda}} e^V e^{-i\Lambda} \quad . \quad (4.2.74)$$

We have defined covariantly chiral superfields Φ by (4.2.64). We can use Ω (see (4.2.69)) to express them in terms of *ordinary* chiral superfields Φ_0 (which we called Φ in sect. 4.2.a):

$$\Phi = e^{\bar{\Omega}} \Phi_0 \quad , \quad \bar{D}_{\dot{\alpha}} \Phi_0 = 0 \quad . \quad (4.2.75)$$

The factor $e^{\bar{\Omega}}$ converts K -transforming fields into Λ -transforming fields:

$$(\Phi_0)' = (e^{-\bar{\Omega}} \Phi)' = e^{i\Lambda} \Phi_0 \quad . \quad (4.2.76)$$

* * *

A useful identity that follows from the explicit form (4.2.68) expresses $\delta \nabla_{\alpha}$ in terms of an arbitrary variation $\delta \Omega$:

$$\delta \nabla_{\alpha} = (\delta e^{-\Omega}) e^{\Omega} \nabla_{\alpha} + \nabla_{\alpha} e^{-\Omega} \delta e^{\Omega} = [\nabla_{\alpha}, e^{-\Omega} \delta e^{\Omega}] \quad . \quad (4.2.77)$$

b.3. Gauge chiral representation

We can also use Ω to go to gauge chiral representation in which *all* quantities are K -inert and transform only under Λ . This is analogous to and not to be confused with the supersymmetry chiral representation (3.3.24-27), (3.4.8). We make a similarity transformation

$$\nabla_{0A} = e^{-\bar{\Omega}} \nabla_A e^{\bar{\Omega}} = (e^{-V} D_{\alpha} e^V, \bar{D}_{\dot{\alpha}}, -i\{\nabla_{0\alpha}, \nabla_{0\dot{\alpha}}\}) \quad ,$$

$$\Phi_0 = e^{-\bar{\Omega}} \Phi \quad ,$$

$$\tilde{\Phi}_0 = \bar{\Phi} e^{\bar{\Omega}} = \overline{(\Phi_0)} e^V \quad . \quad (4.2.78)$$

The quantities ∇_{0A} and Φ_0 are the chiral representation ∇_A and Φ of the previous subsection. We sometimes write the chiral representation hermitian conjugate of Φ_0 as $\tilde{\Phi}_0$ to avoid confusion with the ordinary hermitian conjugate $\bar{\Phi}_0 \equiv \overline{(\Phi_0)}$.

In the chiral representation we see no trace of Ω or K : Only V and Λ appear. However, we necessarily have an asymmetry between chiral and antichiral objects.

c. Bianchi identities

In subsection 4.2.a we analyzed the physical content of the theory using component expansions and the Wess-Zumino gauge. Alternatively, we can find the field content of the theory by “solving” the Bianchi identities. These follow from the Jacobi

identities:

$$[\nabla_{[A}[\nabla_B, \nabla_C]] = 0 \quad , \quad (4.2.79a)$$

which imply

$$\nabla_{[A}F_{BC]} - T_{[AB]}^D F_{D|C]} = 0 \quad . \quad (4.2.79b)$$

Normally these equations are trivial identities. However, once constraints have been imposed on some field strengths, they give information about the remaining ones, and in particular allow one to express all the fields strengths in terms of a basic set. We now describe the procedure.

We solve the equations (4.2.79) subject to the constraints (4.2.60,66) starting with the ones of lowest dimension. For each equation, we consider various pieces irreducible under the Lorentz group, and see what relations are implied among the field strengths. Thus, for example, the relation $[\{\nabla_{(\alpha}, \nabla_{\beta)}\}, \nabla_{\gamma}] = 0$ is identically satisfied when $F_{\alpha,\beta} = 0$. From $[\{\nabla_{\alpha}, \nabla_{\beta}\}, \bar{\nabla}_{\dot{\gamma}}] + [\{\bar{\nabla}_{\dot{\gamma}}, \nabla_{(\alpha}\}, \nabla_{\beta)}] = 0$, we find

$$F_{(\alpha,\beta)\dot{\beta}} = 0 \quad , \quad (4.2.80)$$

which implies, for some spinor superfield W_{β} ,

$$F_{\alpha,\beta\dot{\beta}} = -iC_{\beta\alpha}\bar{W}_{\dot{\beta}} \quad . \quad (4.2.81)$$

From $[\{\nabla_{\alpha}, \nabla_{\beta}\}, \nabla_{\underline{\epsilon}}] + \{[\nabla_{\underline{\epsilon}}, \nabla_{(\alpha}\}, \nabla_{\beta)}] = 0$ we find

$$C_{\gamma(\alpha}\nabla_{\beta)}\bar{W}_{\dot{\gamma}} = 0 \quad , \quad (4.2.82)$$

which implies

$$\nabla_{\alpha}\bar{W}_{\dot{\beta}} = 0 \quad . \quad (4.2.83)$$

From $[\{\nabla_{\alpha}, \bar{\nabla}_{\dot{\beta}}\}, \nabla_{\underline{\epsilon}}] + \{[\nabla_{\underline{\epsilon}}, \nabla_{\alpha}\}, \bar{\nabla}_{\dot{\beta}}\} + \{[\nabla_{\underline{\epsilon}}, \bar{\nabla}_{\dot{\beta}}\}, \nabla_{\alpha}\} = 0$ we obtain

$$F_{\alpha\dot{\beta},\gamma\dot{\gamma}} + C_{\gamma\alpha}\bar{\nabla}_{\dot{\beta}}\bar{W}_{\dot{\gamma}} + C_{\dot{\gamma}\dot{\beta}}\nabla_{\alpha}W_{\gamma} = 0 \quad , \quad (4.2.84)$$

which separates into two equations:

$$F_{\alpha\dot{\alpha},\beta\dot{\beta}} = \frac{1}{2}(C_{\alpha\beta}\bar{\nabla}_{(\dot{\alpha}}\bar{W}_{\dot{\beta})} + C_{\dot{\alpha}\dot{\beta}}\nabla_{(\alpha}W_{\beta)}) \equiv C_{\alpha\beta}\bar{f}_{\dot{\alpha}\dot{\beta}} + C_{\dot{\alpha}\dot{\beta}}f_{\alpha\beta} \quad , \quad (4.2.85)$$

and

$$\nabla^\alpha W_\alpha + \bar{\nabla}^{\dot{\alpha}} \bar{W}_{\dot{\alpha}} = 0 \quad . \quad (4.2.86)$$

These can be reexpressed as

$$\nabla_\alpha W_\beta = i C_{\beta\alpha} D' + f_{\alpha\beta} \quad , \quad D' = \bar{D}' = -\frac{i}{2} \nabla^\alpha W_\alpha \quad . \quad (4.2.87)$$

Finally, $[[\nabla_\alpha, \nabla_{\underline{b}}, \nabla_{\underline{c}}] + [[\nabla_{\underline{b}}, \nabla_{\underline{c}}], \nabla_\alpha] = 0$ and $[[\nabla_{\underline{a}}, \nabla_{\underline{b}}], \nabla_{\underline{c}}] = 0$ are automatically satisfied as a consequence of the previous identities. From (4.2.87) we also obtain

$$\bar{\nabla}_{\dot{\alpha}} D' = \frac{1}{2} \nabla^\beta_{\dot{\alpha}} W_\beta \quad ,$$

$$\bar{\nabla}_{\dot{\alpha}} f_{\alpha\beta} = i \frac{1}{2} \nabla_{(\alpha\dot{\alpha}} W_{\beta)} \quad , \quad (4.2.88)$$

and

$$\nabla_\alpha f_{\beta\gamma} = \frac{1}{2} C_{\alpha(\beta} i \nabla_{\gamma)\dot{\delta}} \bar{W}^{\dot{\delta}} \quad . \quad (4.2.89)$$

Therefore, all the field strengths are expressed in terms of the chiral field strength W^α . In particular, the commutators of the covariant derivatives can be written as:

$$\{\nabla_\alpha, \nabla_\beta\} = 0 \quad ,$$

$$\{\nabla_\alpha, \bar{\nabla}_{\dot{\beta}}\} = i \nabla_{\alpha\dot{\beta}} \quad ,$$

$$[\bar{\nabla}_{\dot{\alpha}}, i \nabla_{\beta\dot{\beta}}] = -i C_{\beta\dot{\alpha}} W_{\dot{\beta}} \quad ,$$

$$[i \nabla_{\underline{a}}, i \nabla_{\underline{b}}] = i (C_{\dot{\alpha}\dot{\beta}} f_{\alpha\beta} + C_{\alpha\beta} \bar{f}_{\dot{\alpha}\dot{\beta}}) \quad . \quad (4.2.90)$$

Furthermore, the set

$$F = \{W_\alpha, D', f_{\alpha\beta}\} \quad , \quad (4.2.91)$$

is closed under the operation of applying ∇_α and $\bar{\nabla}_{\dot{\alpha}}$: Only spacetime derivatives $\nabla_{\underline{a}}$ of F are generated. These superfields are the nonlinear off-shell extension of the superfield strengths $\Psi_{(n)}$ of sec. 3.12. The covariant components are the $\theta = 0$ projections of these superfields. Thus the constraints and the Bianchi identities directly determine the field content of the theory.

4.3. Gauge-invariant models

a. Renormalizable models

In this subsection we consider properties of systems of interacting chiral and real gauge superfields with actions of the form

$$S = \int d^4x d^4\theta \bar{\Phi}_j (e^V)^j_i \Phi^i + tr \int d^4x d^2\theta W^2 + \left[\int d^4x d^2\theta P(\Phi^i) + h.c. \right] \quad (4.3.1)$$

(in the gauge-chiral representation), invariant under a group G . Here $V^i_j = V^A (T_A)^i_j$ and $(T_A)^i_j$ is a (in general reducible) matrix representation of the generators T_A of G . In the vector representation, (4.3.1) takes the form

$$S = \int d^4x d^4\theta \bar{\Phi}_i \Phi^i + tr \int d^4x d^2\theta W^2 + \left[\int d^4x d^2\theta P(\Phi^i) + h.c. \right] \quad (4.3.2)$$

where we have used $tr e^{-\bar{\Omega}} f e^{\bar{\Omega}} = tr f$ in the chiral integral, and rewritten the action in terms of covariantly chiral superfields. The gauge coupling has been set to 1, but can be restored by the rescalings $W_\alpha \rightarrow g^{-1} W_\alpha$. S may be R-symmetric, with the gauge superfield transforming as $V'(x, \theta, \bar{\theta}) = V(x, e^{-ir}\theta, e^{ir}\bar{\theta})$.

Another term can be added to the action: If G is abelian, or has an abelian subgroup, the *Fayet-Iliopoulos* term

$$S_{FI} = tr \int d^4x d^4\theta \nu V = tr \int d^4x \nu D' \quad , \quad (4.3.3)$$

is gauge invariant.

Component actions can be obtained by the projection techniques we have discussed before. A more efficient and, up to field redefinitions, totally equivalent procedure is to define *covariant components* by projecting with covariant derivatives. Thus, for a covariantly chiral superfield we define

$$A = \Phi| \quad , \quad \psi_\alpha = \nabla_\alpha \Phi| \quad , \quad F = \nabla^2 \Phi| \quad . \quad (4.3.4)$$

Similarly, the covariant components of the gauge multiplet can be obtained by projection from W_α (here $f_{\alpha\beta}$ denotes the component field strength):

$$\lambda_\alpha = W_\alpha| \quad , \quad f_{\alpha\beta} = \frac{1}{2} \{ \nabla_{(\alpha} , W_{\beta)} \}| \quad ,$$